Assignment 20.5

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s) Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju		
CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week10 – Friday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
Assignment Number:20.5(Present assignment number)/**24**(Total number of assignments)			

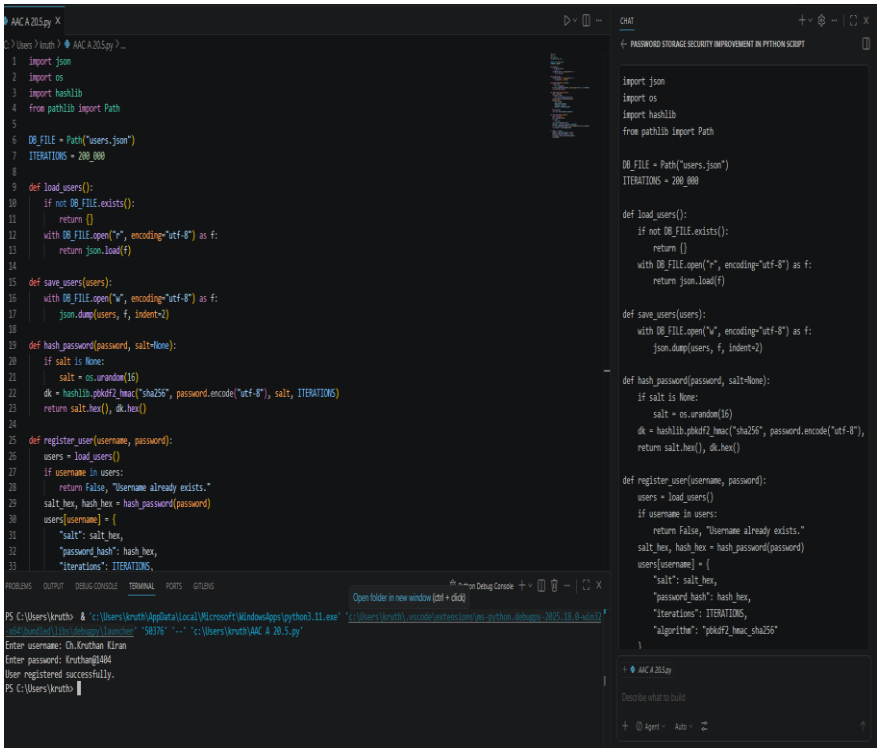
Q.No.	Question	*Expected Time to complete*
1	**Lab 20 – Security Testing: Identifying Vulnerabilities in AI-Generated Code**	Week10 - Friday

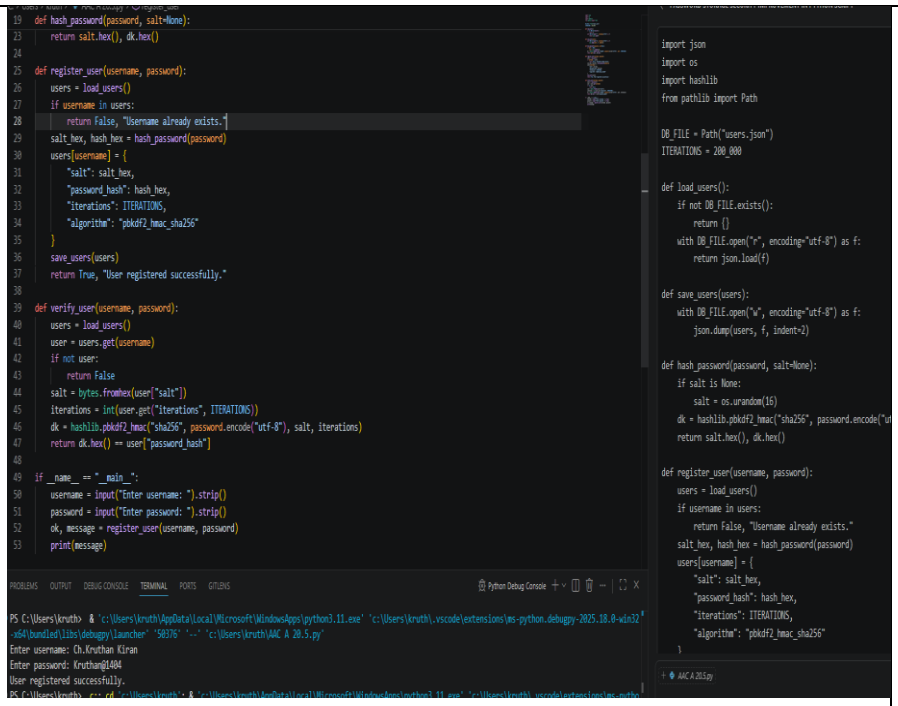
•

Task 1 – Password Storage Vulnerability Check

Task:

Analyze an AI-generated signup script for insecure password storage.





```python
def hash_password(password, salt=None):
    return salt.hex(), dk.hex()

def register_user(username, password):
    users = load_users()
    if username in users:
        return False, "Username already exists."
    salt_hex, hash_hex = hash_password(password)
    users[username] = {
        "salt": salt_hex,
        "password_hash": hash_hex,
        "iterations": ITERATIONS,
        "algorithm": "pbkdf2_hmac_sha256"
    }
    save_users(users)
    return True, "User registered successfully."

def verify_user(username, password):
    users = load_users()
    user = users.get(username)
    if not user:
        return False
    salt = bytes.fromhex(user["salt"])
    iterations = int(user.get("iterations", ITERATIONS))
    dk = hashlib.pbkdf2_hmac("sha256", password.encode("utf-8"), salt, iterations)
    return dk.hex() == user["password_hash"]

if __name__ == "__main__":
    username = input("Enter username: ").strip()
    password = input("Enter password: ").strip()
    ok, message = register_user(username, password)
    print(message)
```

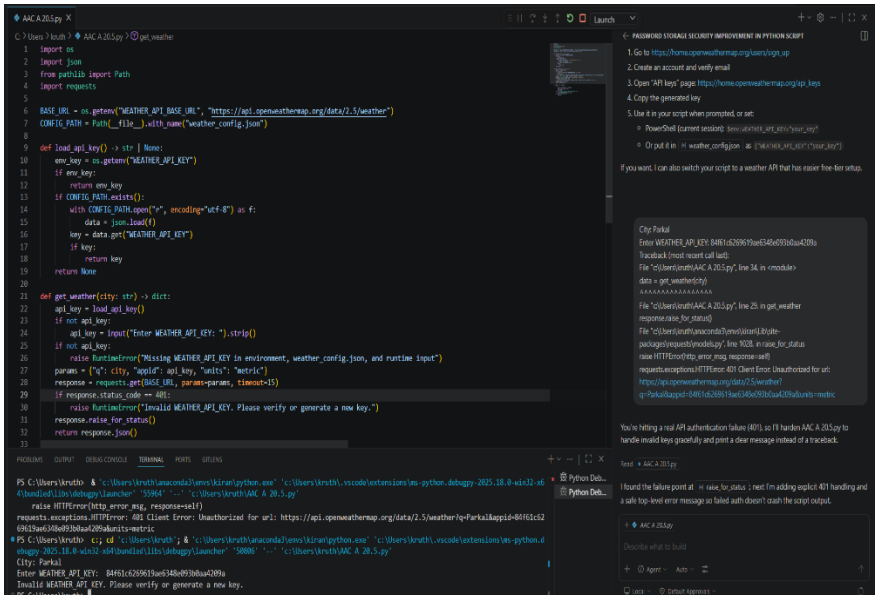
```
PS C:\Users\kruth> & "c:\Users\kruth\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "58376" "--" "c:\Users\kruth\AAC A 20.5.py"
Enter username: Ch.Kruthan Kiran
Enter password: Kruthan@1404
User registered successfully.
```

```python
import json
import os
import hashlib
from pathlib import Path

DB_FILE = Path("users.json")
ITERATIONS = 200_000

def load_users():
    if not DB_FILE.exists():
        return {}
    with DB_FILE.open("r", encoding="utf-8") as f:
        return json.load(f)

def save_users(users):
    with DB_FILE.open("w", encoding="utf-8") as f:
        json.dump(users, f, indent=2)

def hash_password(password, salt=None):
    if salt is None:
        salt = os.urandom(16)
    dk = hashlib.pbkdf2_hmac("sha256", password.encode("ut
    return salt.hex(), dk.hex()

def register_user(username, password):
    users = load_users()
    if username in users:
        return False, "Username already exists."
    salt_hex, hash_hex = hash_password(password)
    users[username] = {
        "salt": salt_hex,
        "password_hash": hash_hex,
        "iterations": ITERATIONS,
        "algorithm": "pbkdf2_hmac_sha256"
    }
```

Task 2 – Hardcoded Secrets Detection

Task:

Evaluate AI-generated code for hardcoded API keys or credentials.



```python
import os
import json
from pathlib import Path
import requests

BASE_URL = os.getenv("WEATHER_API_BASE_URL", "https://api.openweathermap.org/data/2.5/weather")
CONFIG_PATH = Path(__file__).with_name("weather_config.json")

def load_api_key() -> str | None:
    env_key = os.getenv("WEATHER_API_KEY")
    if env_key:
        return env_key
    if CONFIG_PATH.exists():
        with CONFIG_PATH.open("r", encoding="utf-8") as f:
            data = json.load(f)
        key = data.get("WEATHER_API_KEY")
        if key:
            return key
    return None

def get_weather(city: str) -> dict:
    api_key = load_api_key()
    if not api_key:
        api_key = input("Enter WEATHER_API_KEY: ").strip()
    if not api_key:
        raise RuntimeError("Missing WEATHER_API_KEY in environment, weather_config.json, and runtime input")
    params = {"q": city, "appid": api_key, "units": "metric"}
    response = requests.get(BASE URL, params=params, timeout=15)
    if response.status_code == 401:
        raise RuntimeError("Invalid WEATHER_API_KEY. Please verify or generate a new key.")
    response.raise_for_status()
    return response.json()
```

```
PS C:\Users\kruth> & "c:\Users\kruth\anaconda3\envs\kiran\python.exe" "c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "55964" "--" "c:\Users\kruth\AAC A 20.5.py"
    raise HTTPError(http_error_msg, response=self)
requests.exceptions.HTTPError: 401 Client Error: Unauthorized for url: https://api.openweathermap.org/data/2.5/weather?q=Parkal&appid=84f61c626913ae6348e093b0aa4209a&units=metric
PS C:\Users\kruth> c; cd "c:\Users\kruth"; & "c:\Users\kruth\anaconda3\envs\kiran\python.exe" "c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "58006" "--" "c:\Users\kruth\AAC A 20.5.py"
City: Parkal
Enter WEATHER_API_KEY:  84f61c626913ae6348e093b0aa4209a
Invalid WEATHER_API_KEY. Please verify or generate a new key.
```

PASSWORD STORAGE SECURITY IMPROVEMENT IN PYTHON SCRIPT

1. Go to https://home.openweathermap.org/users/sign_up
2. Create an account and verify email
3. Open "API keys" page: https://home.openweathermap.org/api_keys
4. Copy the generated key
5. Use it in your script when prompted, or set:
   - PowerShell (current session): `$env:WEATHER_API_KEY="your_key"`
   - Or put it in `weather_config.json` as `{"WEATHER_API_KEY":"your_key"}`

If you want, I can also switch your script to a weather API that has easier free-tier setup.

```
City: Parkal
Enter WEATHER_API_KEY: 84f61c626913ae6348e093b0aa4209a
Traceback (most recent call last):
  File "c:\Users\kruth\AAC A 20.5.py", line 34, in <module>
    data = get_weather(city)
           ^^^^^^^^^^^^^^^^^
  File "c:\Users\kruth\AAC A 20.5.py", line 25, in get_weather
    response.raise_for_status()
  File "c:\Users\kruth\anaconda3\envs\kiran\Lib\site-packages\requests\models.py", line 1028, in raise_for_status
    raise HTTPError(http_error_msg, response=self)
requests.exceptions.HTTPError: 401 Client Error: Unauthorized for url: https://api.openweathermap.org/data/2.5/weather?q=Parkal&appid=84f61c626913ae6348e093b0aa4209a&units=metric
```

You're hitting a real API authentication failure (401), so I'll harden AAC A 20.5.py to handle invalid keys gracefully and print a clear message instead of a traceback.

Read AAC A 20.5.py

I found the failure point at `response.raise_for_status()`; next I'm adding explicit 401 handling and a safe top-level error message so failed auth doesn't crash the script output.

AAC A 20.5.py

Describe what to build

Agent · Auto

Local · Default approvals

AAC A 20.5.py

C: > Users > kruth > AAC A 20.5.py > get_weather

```python
def get_weather(city: str) -> dict:
    api_key = load_api_key()
    if not api_key:
        api_key = input("Enter WEATHER_API_KEY: ").strip()
    if not api_key:
        raise RuntimeError("Missing WEATHER_API_KEY in environment, weather_config.json, and runtime input")
    params = {"q": city, "appid": api_key, "units": "metric"}
    response = requests.get(BASE_URL, params=params, timeout=15)
    if response.status_code == 401:
        raise RuntimeError("Invalid WEATHER_API_KEY. Please verify or generate a new key.")
    response.raise_for_status()
    return response.json()

if __name__ == "__main__":
    try:
        city = input("City: ").strip()
        data = get_weather(city)
        print(data)
    except requests.exceptions.RequestException as e:
        print(f"Network/API error: {e}")
    except RuntimeError as e:
        print(e)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS C:\Users\kruth>  & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55964' '--' 'c:\Users\kruth\AAC A 20.5.py'
    raise HTTPError(http_error_msg, response=self)
requests.exceptions.HTTPError: 401 Client Error: Unauthorized for url: https://api.openweathermap.org/data/2.5/weather?q=Parkal&appid=84f61c6269619ae6348e093b0aa4209a&units=metric
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50806' '--' 'c:\Users\kruth\AAC A 20.5.py'
City: Parkal
Enter WEATHER_API_KEY:  84f61c6269619ae6348e093b0aa4209a
Invalid WEATHER_API_KEY. Please verify or generate a new key.
```

Task 3 – File Upload Security Validation

Task:

Test an AI-generated file upload program for unsafe file handling.

AAC A 20.5.py 2

```python
from flask import Flask, request, jsonify
from werkzeug.utils import secure_filename
from pathlib import Path
import uuid

app = Flask(__name__)
app.config["MAX_CONTENT_LENGTH"] = 5 * 1024 * 1024

UPLOAD_DIR = Path("uploads")
UPLOAD_DIR.mkdir(parents=True, exist_ok=True)

ALLOWED_EXTENSIONS = {"png", "jpg", "jpeg", "pdf", "txt"}
EXTENSION_TO_KIND = {
    "png": {"png"},
    "jpg": {"jpeg"},
    "jpeg": {"jpeg"},
    "pdf": {"pdf"},
    "txt": {"txt"},
}

def detect_kind(data: bytes) -> str:
    if data.startswith(b"\x89PNG\r\n\x1a\n"):
        return "png"
    if data.startswith(b"\xff\xd8\xff"):
        return "jpeg"
    if data.startswith(b"%PDF-"):
        return "pdf"
    if data.startswith(b"MZ"):
        return "exe"
    if data.startswith(b"PK\x03\x04"):
        return "zip"
    if b"\x00" in data:
        return "binary"
    try:
        data.decode("utf 8")
```

PASSWORD STORAGE SECURITY IMPROVEMENT IN PYTHON SCRIPT

```
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57510' '--' 'c:\Users\kruth\AAC A 20.5.py'
ModuleNotFoundError: No module named 'flask'
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57510' '--' 'c:\Users\kruth\AAC A 20.5.py'
 * Serving Flask app 'AAC A 20.5'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
PS C:\Users\kruth>
```

AAC A 20.5.py 2

C: > Users > kruth > AAC A 20.5.py > ...

```python
def detect_kind(data: bytes) -> str:
    try:
        data.decode("utf-8")
        return "txt"
    except UnicodeDecodeError:
        return "unknown"

@app.post("/upload")
def upload_file():
    if "file" not in request.files:
        return jsonify({"error": "No file part"}), 400

    file = request.files["file"]
    if file.filename is None or file.filename.strip() == "":
        return jsonify({"error": "No selected file"}), 400

    original_name = secure_filename(file.filename)
    if original_name == "":
        return jsonify({"error": "Invalid filename"}), 400

    ext = Path(original_name).suffix.lower().lstrip(".")
    if ext not in ALLOWED_EXTENSIONS:
        return jsonify({"error": "File extension not allowed"}), 400

    head = file.stream.read(8192)
    if not head:
        return jsonify({"error": "Empty file"}), 400

    kind = detect_kind(head)
    if kind in {"exe", "zip", "binary", "unknown"}:
        return jsonify({"error": "Unsafe or unsupported file content"}), 400

    if kind not in EXTENSION_TO_KIND[ext]:
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-
ebugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57474' '--' 'c:\Users\kruth\AAC A 20.5.py'
ModuleNotFoundError: No module named 'flask'
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-
ebugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57510' '--' 'c:\Users\kruth\AAC A 20.5.py'
 * Serving Flask app 'AAC A 20.5'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

AAC A 20.5.py 2

C: > Users > kruth > AAC A 20.5.py > ...

```python
def upload_file():
    if original_name == "":
        return jsonify({"error": "Invalid filename"}), 400

    ext = Path(original_name).suffix.lower().lstrip(".")
    if ext not in ALLOWED_EXTENSIONS:
        return jsonify({"error": "File extension not allowed"}), 400

    head = file.stream.read(8192)
    if not head:
        return jsonify({"error": "Empty file"}), 400

    kind = detect_kind(head)
    if kind in {"exe", "zip", "binary", "unknown"}:
        return jsonify({"error": "Unsafe or unsupported file content"}), 400

    if kind not in EXTENSION_TO_KIND[ext]:
        return jsonify({"error": "File content does not match extension"}), 400

    file.stream.seek(0)
    stored_name = f"{uuid.uuid4().hex}_{original_name}"
    destination = UPLOAD_DIR / stored_name
    file.save(destination)

    return jsonify({"message": "Upload successful", "file": stored_name}), 201

@app.errorhandler(413)
def file_too_large(_):
    return jsonify({"error": "File too large. Max size is 5 MB"}), 413

if __name__ == "__main__":
    app.run(debug=False)
```

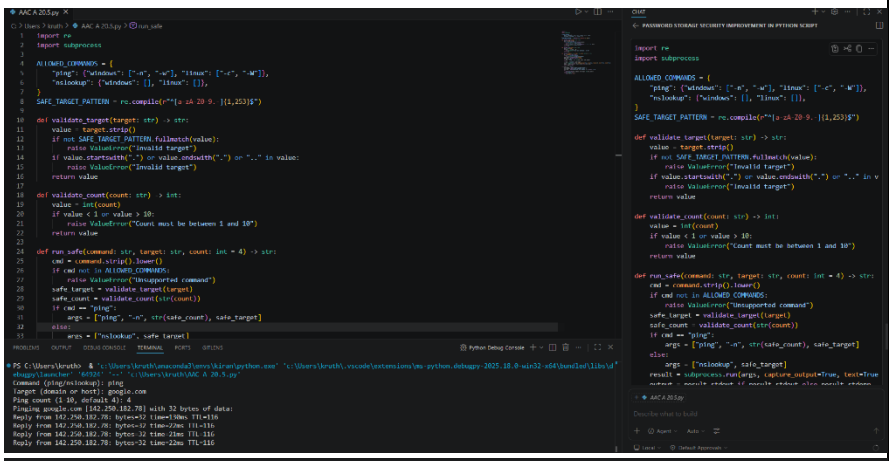
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

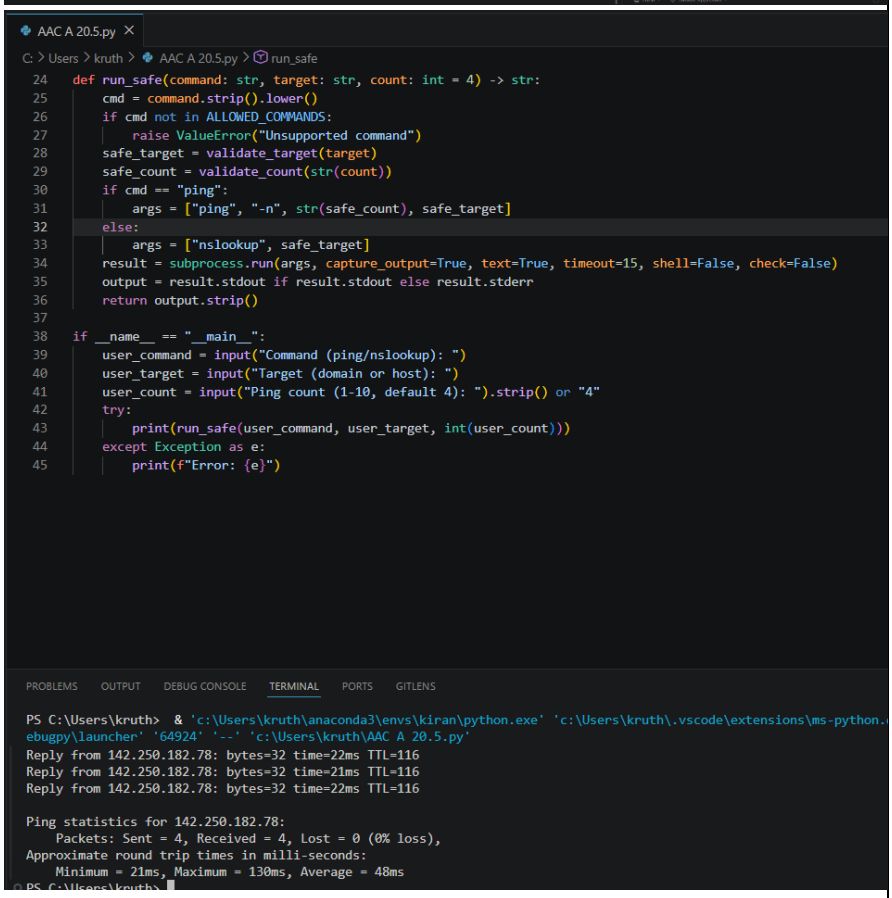
```
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\ex
ebugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57474' '--' 'c:\Users\kruth\AAC A 20.5.py'
ModuleNotFoundError: No module named 'flask'
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\ex
ebugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57510' '--' 'c:\Users\kruth\AAC A 20.5.py'
 * Serving Flask app 'AAC A 20.5'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

Task 4 – Command Injection Prevention

Task:

Analyze an AI-generated Python script that executes system commands.

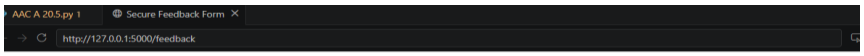




Task 5 – Secure Input Handling in Web Forms

Task:

Check AI-generated web form code for improper input validation.



Feedback

Name

Kruthan Kiran

Email

kruthankiran@gmail.com

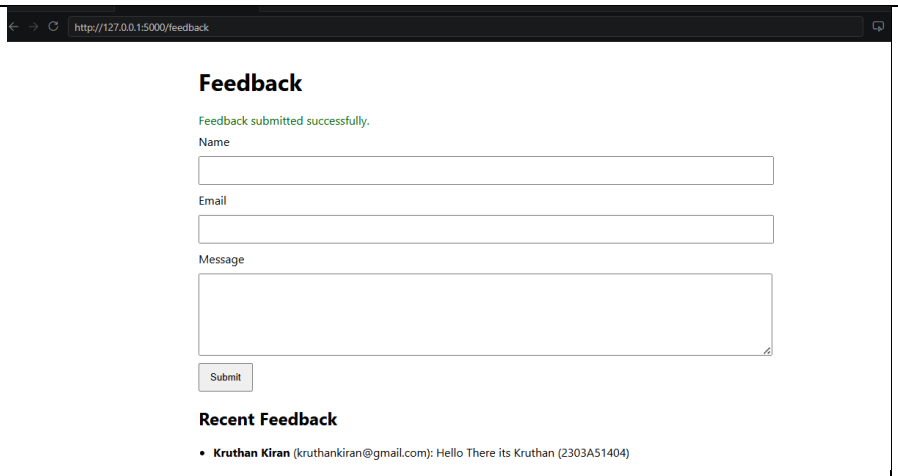
Message

Hello There its Kruthan (2303A51404)

Submit

Recent Feedback

- No feedback yet.



Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.